BAHRIA University, Islamabad
Center of Excellence – Artificial Intelligence

Pedoscan Sensor Report

By
Ghufran Ali Khokhar
(Software Developer)

To
Ms. Amna
(Instructor)

Table of Contents

1. Introduction.....3
2. Aim and Objective.....3
3. Literature Review.....4
4. Types of Sensors Used.....4
   4.1 FSR (Force Sensitive Resistor).....4
   4.2 Piezoelectric Sensors.....4
   4.3 Load Cell Array (Strain Gauge Type).....5
   4.4 Velostat Matrix (Pressure-Sensitive Conductive Sheet).....5
5. Preferred Sensor.....6
   5.1 Description.....7
   5.2 Features and Benefits.....8
   5.3 Applications.....8
6. Conclusion.....9

1. Introduction

Pedoscan is a foot function assessment system that measures the pressure distribution of a person's foot when weight is applied. It provides valuable information about foot shape, balance, gait, and load distribution, important parameters for identifying abnormalities, especially in diabetic patients (neuropathy cases) who are prone to foot wounds or deformities.

Figure 1

Pedoscan systems help clinicians analyze:

- Balance and weight distribution
- Pressure mapping and gait analysis
- Center of force and sway analysis

By capturing real-time pressure data, the Pedoscan assists in diagnosing irregularities such as high-pressure zones, improper weight transfer, or abnormal foot posture.

2. Aim and Objective

To identify and select the most compatible sensor for the Pedoscan system that provides accurate foot pressure measurement, balance assessment, and gait analysis while maintaining cost-effectiveness and ease of integration.

3. Literature Review

Pedoscan systems are used for analyzing foot pressure distribution and gait patterns. They play an important role in detecting foot deformities, balance issues, and pressure-related injuries, especially in diabetic patients. Various sensor technologies such as FSR, capacitive, piezoelectric, and load cell arrays have been explored for pressure mapping. Among these, FSR-based systems are the most common due to their simplicity, flexibility, and cost-effectiveness, making them suitable for both clinical and research applications.

4. Types of Sensors Used

Different types of sensors are utilized in Pedoscan systems to measure foot pressure and gait characteristics. Each sensor type has unique advantages depending on accuracy, cost, and application needs.

4.1. FSR (Force Sensitive Resistor)

FSRs change their resistance based on the applied pressure. They are lightweight, flexible, and easy to interface with microcontrollers, making them ideal for low-cost foot pressure mapping and Pedoscan systems.

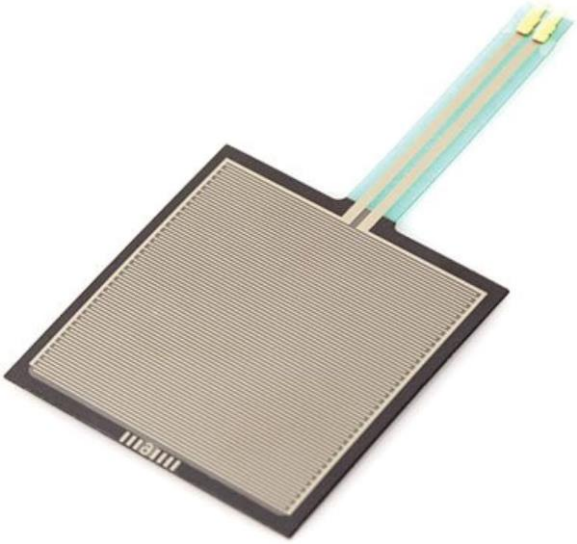

Figure 2

4.2. Piezoelectric Sensors

Piezoelectric materials generate a voltage when subjected to mechanical stress. They are suitable for measuring dynamic forces such as walking or running patterns but are less effective for static pressure analysis.

Figure 3

4.3. Load Cell Array (Strain Gauge Type)

Load cells use strain gauges to convert force into electrical signals. They provide highly accurate measurements and are commonly used in weight and force applications, though they are bulkier and less flexible for wearable systems.

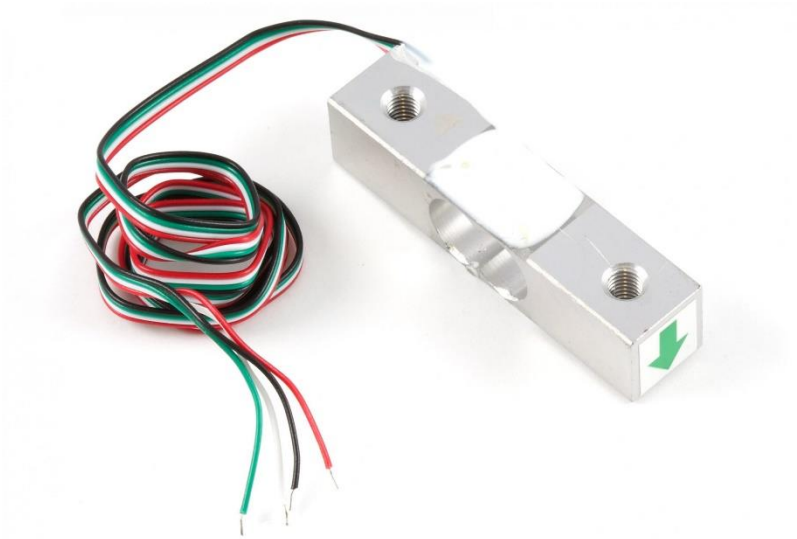

Figure 4

4.4. Velostat Matrix (Pressure-Sensitive Conductive Sheet)

Velostat is a low-cost, flexible material whose resistance changes with pressure. It is widely used in DIY or research-level Pedoscan systems for building pressure-sensitive mats with customizable resolutions.

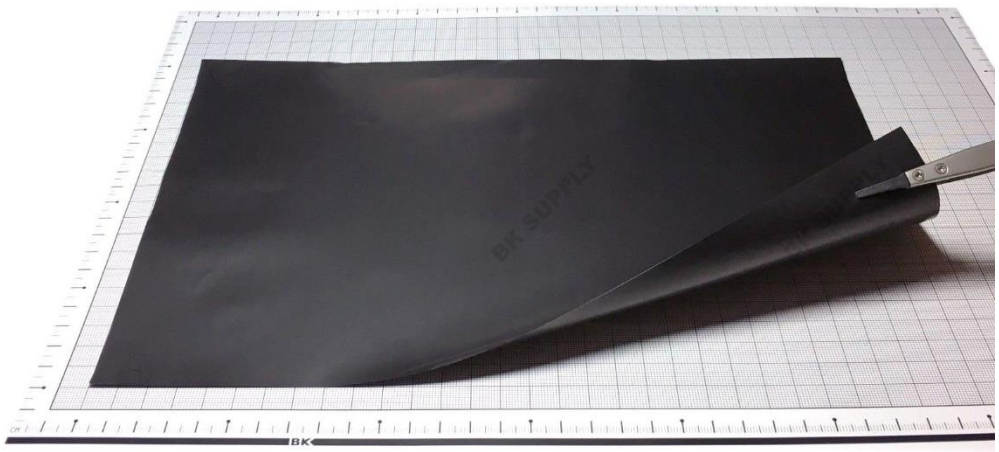

Figure 5

5. Preferred Sensor

The Matrix Force Sensing Resistor (FSR 60×60) is preferred as the most compatible sensor for the Pedoscan system due to its high resolution, fast response, thin flexible design, and cost-effective. It is an ultra-sensitive human body pressure mapping sensor ideal for continuous force measurement and pressure distribution analysis.

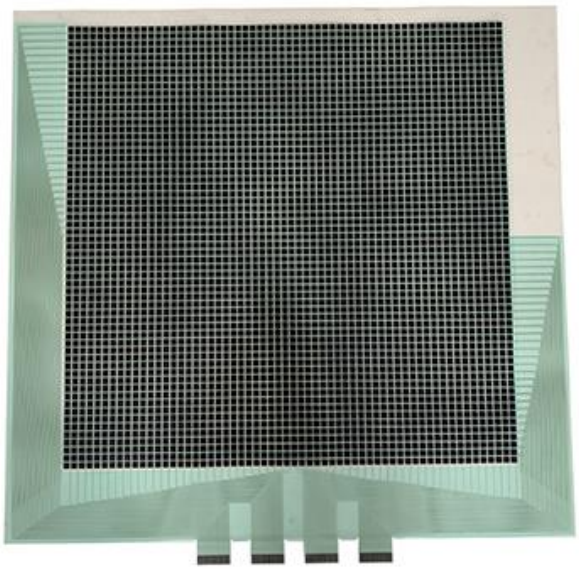

Figure 6: 60x60 Matric FSR

Model	Matrix FSR 60×60
Type	Multipoint / Matrix Force Sensor
Sensing Area	418.5 × 418.5 mm
Overall Dimension	473.28 × 525.00 mm
Force Range	200 g – 100 kg
Thickness	≤ 0.3 mm
Response Time	< 5 ms
Life Span	> 2 million cycles
Connector	ZIF 1.0 mm (120 pins)
Operating Temperature	–20°C to 80°C

Table 1

5.1. Description

The Matrix FSR operates on a grid structure composed of 60 rows and 60 columns, forming 3600 sensing nodes (intersection points). Each node acts as an independent sensor that changes resistance according to the applied pressure.

When pressure is applied, nodes with higher force show lower resistance and higher output voltage, allowing the system to generate a real-time pressure distribution map across the foot. This makes it ideal for precise pressure mapping, gait analysis, and balance evaluation in Pedoscan applications.

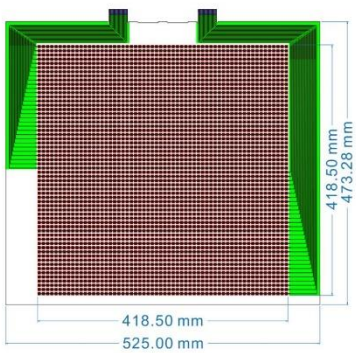

Figure 7

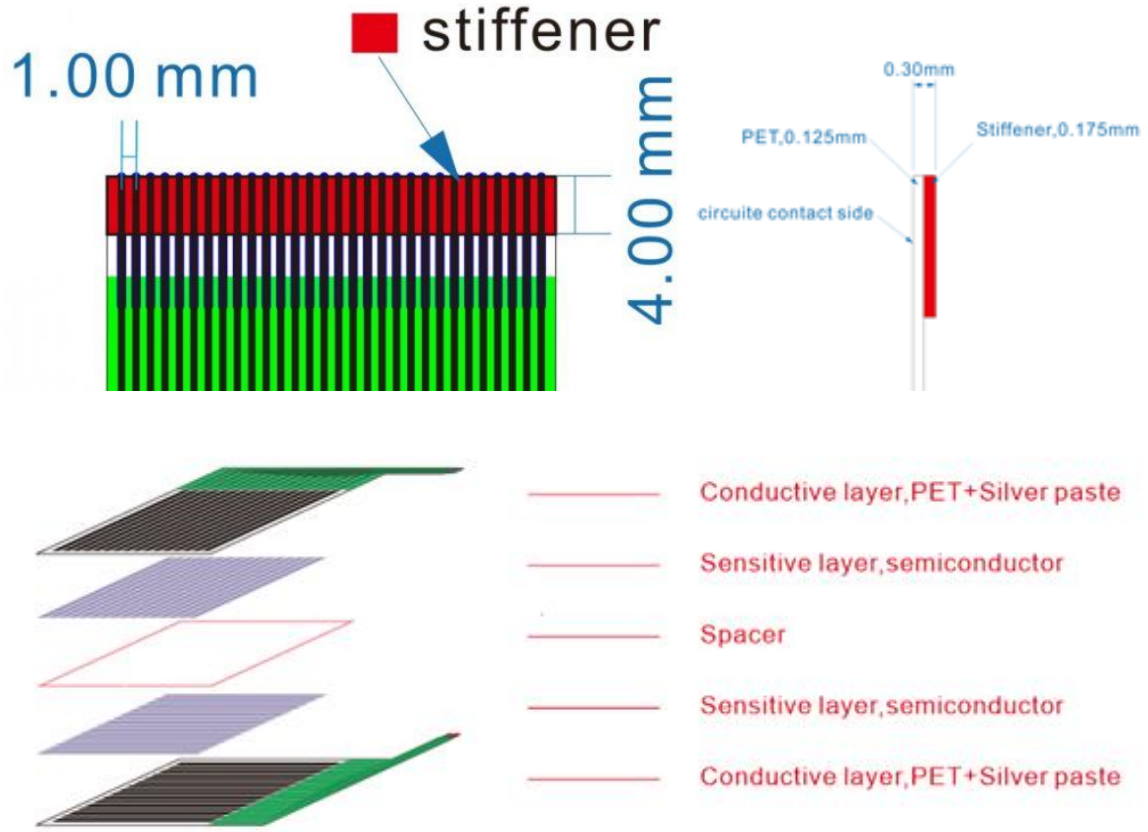

Figure 8

5.2. Features and Benefits

- Detects forces from as low as 200 g up to 100 kg.
- Ultra-thin and flexible, easy to integrate under foot platforms.
- Fast response (<5 ms) enables real-time measurement.
- Provides continuous force resolution for detailed mapping.
- Cost-effective solution compared to capacitive or piezoelectric sensors.

5.3. Applications

This sensor is versatile and widely used in:

- **Medical and Biomechanical Analysis:** Foot pressure mapping, gait and balance assessment, prosthetics, and rehabilitation.
- **Sports & Fitness:** Performance and posture evaluation.
- **Robotics and Embedded Systems:** Touch and grip force sensing.
- **Automotive and Industrial Testing:** Machine alignment and nip pressure analysis.

6. Conclusion

In conclusion, the Matrix FSR 60×60 is the most suitable choice for the Pedoscan system because it offers high spatial resolution (3600 sensing points), flexibility, and affordability, making it perfect for accurate foot pressure mapping and gait evaluation. Its ultra-thin design allows easy integration into measurement platforms, ensuring comfort and reliability during testing. The fast response time and wide force range enable precise detection of subtle pressure variations under the foot. Overall, this sensor provides an efficient, durable, and cost-effective solution for developing an advanced foot pressure analysis system.